

MIXING-GATED DIFFUSION: BUDGETED, PER-NODE, PER-HOP CONTROL FOR OVERSMOOTHING-RESILIENT GRAPH LEARNING

Anonymous authors

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ABSTRACT

Oversmoothing arises when cross-group mixing outpaces within-group denoising in graph learning, flattening node representations and hurting accuracy. Non-asymptotic theory shows this trade-off is local across nodes and hops, while recent analyses prove that attention by itself cannot avert exponential loss of expressivity with depth. Existing remedies either apply global filters, normalize post hoc, or inject randomness, and none yields a deterministic, interpretable policy that decides which hops each node should aggregate. We propose MiGAD, a budgeted, compatibility-aware diffusion guard. MiGAD learns soft groups and a row-stochastic group compatibility matrix, estimates per-node, per-hop mixing risk by contrasting local compatible groups with k -hop group diffusion, and enforces a node-specific budget via a differentiable, constrained hop-attention layer. The method plugs into decoupled multi-hop backbones, remains complementary to normalization, and provides deterministic, per-node hop allocations at inference. We specify end-to-end PyTorch/Geometric code, diagnostics, ablations, and rigorous protocols spanning depth robustness, heterophily and corruption, and interpretability. Our provided run establishes an executable environment and training pipeline; quantitative outcomes were not captured in the shared logs, so we report no numbers, while supplying reproducible procedures to test selective long-range diffusion.

1 INTRODUCTION

Graph neural networks succeed when aggregation denoises within-class variability without erasing the structure that distinguishes classes. A non-asymptotic analysis formalizes this balance as a local competition between desirable within-group denoising and harmful cross-group mixing, predicting oversmoothing once the latter dominates at a depth that depends on node and hop rather than a global layer count Wu et al. (2022). At the same time, rigorous results show that attention mechanisms cannot, on their own, prevent exponential loss of expressive power with depth, even when aggregation is asymmetric, state-dependent, and time-varying Wu et al. (2023). These insights motivate a guardrail that constrains aggregation itself, instead of relying on global filters or repairing representations only after aggregation.

Current practice offers several partial fixes. Spectral designs introduce graph-adapted filters but do not personalize depth across nodes or hops [bruna2013spectral, deferrard2016convolutional]. Hop — *decoupled pipelines precompute multi-hop features and then fuse them, improving scalability yet remaining risk-agnostic at fusion time* Chen et al. (2022). Node-dependent depth selector sets a single cutoff per node based on graph-only stationary proximity, averaging over all earlier hops and ignoring feature and label compatibility Zhan et al. (2021) and from which hops to aggregate [zhao2019pairnorm, zhou2020towards, guo2023contranorm]. Stochastic strategies

We introduce MiGAD, a Mixing-GAted Diffusion layer that operationalizes the local mixing-denoising trade-off. MiGAD learns a compact soft grouping S of nodes and a low-rank, row-stochastic compatibility matrix C between groups. It propagates group memberships across hops, computes a per-node, per-hop mixing risk by comparing the locally compatible groups $S_i C$ to the k -hop group mix $M_i^{(k)} = P^k S$, and pairs this with a denoising proxy that rewards hop-wise stability.

Crucially, MiGAD imposes a node-specific budget ρ_i and solves a constrained hop-attention problem that maximizes utility under the constraint that the expected risky mass—the sum over hops of weights times risks—does not exceed ρ_i . A differentiable per-node dual variable computes deterministic hop allocations at inference.

1.1 CHALLENGES IN REGULATING HOP-WISE AGGREGATION

Safe aggregation depths vary per node and per hop. Graph-only selectors do not assess compatibility with features or labels, cannot gate individual hops, and average earlier layers uniformly Zhang et al. (2021). Hop-decoupled attention fuses without a theory-aligned local constraint. Normalizers repair but do not regulate what to aggregate. Stochastic methods trade determinism for robustness and yield no interpretable inference-time policy [rong2019_droppedge, papp2021_droppedggn]. Meanwhile, attention's depth limits and global filters' coarse control leave a gap for conditional, hop-wise governance [wu2023_demytifying, bruna2013_spectral, deferrard2016_convolutional].

1.2 PROPOSED APPROACH AND POSITIONING

Our approach addresses these gaps with a budgeted, compatibility-aware gating mechanism that is deterministic, interpretable, and plug-and-play. It aligns with non-asymptotic theory by constraining the accumulation of risky mass at the hop level Wu et al. (2022). It also fits neatly into hop-interaction pipelines that decouple diffusion from learning Chen et al. (2022), and remains complementary to PairNorm, DGN, and ContraNorm [zhao2019_pairnorm, zhou2020_towards, guo2023_contranorm].

1.3 CONTRIBUTIONS

- **Compatibility-aware risk and budgets.** We formulate a compatibility-aware, per-node, per-hop mixing risk from learned soft groups and a learnable compatibility matrix, and we impose adaptive node budgets that cap risky mass.
- **Constrained hop attention.** We derive a differentiable, per-node constrained hop-attention layer with a single dual variable per node that enforces risk budgets while maximizing utility.
- **Practical implementation.** We provide a practical, scalable PyTorch Geometric implementation with group sparsity, random-walk risk estimation, and deterministic inference; the module drops into SIGN, SGC, GPR, APPNP, and hop-interaction backbones Chen et al. (2022).
- **Rigorous protocols.** We design rigorous, reproducible protocols and diagnostics for depth robustness, oversmoothing control, calibration, heterophily and corruption, and interpretability.

1.4 VERIFICATION PLAN AND BROADER SCOPE

Verification plan. We specify three experiments: depth-accuracy area under curve and oversmoothing diagnostics on Planetoid and synthetic stochastic block models; robustness under heterophily and corruption of edges, features, and labels; and interpretability via boundary correlation, compatibility recovery, and counterfactual risky-hop enablement. Baselines include SIGN (uniform and attention), SGC, APPNP, GPR-GNN, NDLS, normalization layers, and stochastic methods [zhao2019_pairnorm, zhou2020_towards, guo2023_contranorm, rong2019_droppedge, papp2021_droppedggn, zhang2021_node, chen2021_hop].

Beyond node classification, the budgeted-risk principle extends naturally to graph transformers by budgeting long-range attention, to temporal and heterogeneous graphs via time- and type-conditioned compatibilities, and to data governance tasks such as active learning and rewiring curricula. We situate MiGAD alongside universality perspectives and inductive knowledge graph completion to clarify scope and non-applicability [author_year_universal, author_year_indigo].

2 RELATED WORK

2.1 OVERSMOOTHING THEORY AND SPECTRAL FOUNDATIONS

Spectral networks provide graph-adapted convolutional filters and efficient localized approximations, yet they are configured globally and do not tailor aggregation depth to local node-hop conditions [bruna₂₀₁₃_{spectral}, deferrard₂₀₁₆_{convolutional}]. Anon – asymptotic analysis quantifies that oversmoothing arises once cross-group mixing dominates within-group denoising, with a depth that depends on local structure rather than a fixed global threshold.

2.2 ATTENTION-BASED MODELS

Despite their flexibility, attention-based GNNs cannot avert exponential loss of expressivity with depth, even when attention varies across time and states Wu et al. (2023). Expressive variants such as GATv2 strengthen the attention mechanism but do not change this fundamental depth limitation Brody et al. (2021). MiGAD therefore treats attention as a fusion mechanism that must be constrained by a local, theory-aligned budget to prevent unsafe mixing.

2.3 DECOUPLED AND HOP-INTERACTION ARCHITECTURES

Hop-decoupled pipelines precompute multi-hop features $H^{(k)} = P^k X$ and then fuse them via fixed or learned weights, improving efficiency and stability Chen et al. (2022). However, these fusions are risk-agnostic: they do not enforce node-conditional constraints that cap risky mass per hop. MiGAD is architecturally compatible with these pipelines and replaces the fusion stage with a budgeted layer that produces deterministic, per-node hop policies.

2.4 DEPTH SELECTION AND SMOOTHING CONTROL

Node Dependent Local Smoothing personalizes smoothing by selecting a node-specific iteration from graph-only influence scores, then averaging all earlier hops uniformly Zhang et al. (2021). NDLS does not exploit features or labels to assess compatibility, cannot gate individual hops, and cannot learn per-hop trade-offs. MiGAD learns soft groups and a compatibility matrix to produce feature- and label-aware risks, and it enforces a per-node budget at the hop level, yielding interpretable inference-time decisions.

2.5 NORMALIZATION AND POST-HOC REMEDIES

PairNorm, DGN, and ContraNorm reduce oversmoothing symptoms by reshaping embeddings to avoid collapse or dimensional narrowing [zhao₂₀₁₉_{pairnorm}, zhou₂₀₂₀_{towards}, guo₂₀₂₃_{contranorm}]. These are complementary to MiGAD: they influence representation geometry but do not decide which hops to aggregate. MiGAD modulates aggregation directly.

2.6 STOCHASTIC STRATEGIES AND EXPRESSIVENESS

DropEdge randomly removes edges per epoch to slow oversmoothing and reduce overfitting Rong et al. (2019), while DropGNN executes multiple dropout-perturbed runs to increase expressiveness Papp et al. (2021). These methods improve training but offer no deterministic, interpretable inference policies. MiGAD provides a reproducible hop allocation governed by a per-node budget.

2.7 UNIVERSALITY AND OTHER DOMAINS

Results on universal graph convolution concern expressivity guarantees rather than local, risk-constrained diffusion [author_{year}_{universal}]. Inductive knowledge graph completion targets relational reasoning with different supervision and gated diffusion, and we do not treat them as baselines.

3 BACKGROUND

3.1 PROBLEM SETTING

Let G be a graph with N nodes, edges E , and node features $X \in \mathbb{R}^{N \times d}$. Let P be a row-stochastic normalization of the adjacency with self-loops so that each row sums to one. In a decoupled multi-hop pipeline, we precompute $H^{(k)} = P^k X$ for hops $k \in \{0, \dots, L\}$ and later fuse these representations for prediction. Oversmoothing emerges when, for a given node and hop, the cross-group mixing induced by P^k outweighs the within-group denoising, a transition predicted to occur at depths that depend on local structure Wu et al. (2022).

3.2 SOFT GROUPS AND COMPATIBILITY

We posit latent soft groups captured by $S \in \mathbb{R}^{N \times G}$, where each row S_i is a distribution over G groups produced by a small head applied to X . To avoid trivial homophily and collapse, we regularize row entropies and anchor S with prototypes initialized by K-means over X . Inter-group compatibility is a row-stochastic matrix $C \in \mathbb{R}^{G \times G}$, parameterized as $C = \text{softmax}_{\text{row}}(UV^\top)$ with a low-rank factorization and mild priors encouraging near-identity while allowing off-diagonal flexibility.

3.3 HOP-WISE MIXING RISK

We diffuse S across the graph as $M^{(k)} = P^k S$. For node i and hop k , we compare $S_i C$, the distribution of groups deemed compatible with i , to $M_i^{(k)}$, the k -hop mix of groups around i . A base alignment $a_i^{(k)}$ equals the inner product $\langle S_i C, M_i^{(k)} \rangle$, clipped to $[0, 1]$. The base risk is $r_i^{(k)} = 1 - a_i^{(k)}$. Robust alternatives use Jensen-Shannon divergence between $S_i C$ and $M_i^{(k)}$, or one minus their cosine similarity, rescaled to $[0, 1]$. This yields a per-node, per-hop ledger of mixing risks.

3.4 DENOISING PROXY

To promote aggregation that stabilizes representations, we define $d_i^{(k)}$ from the norm of the difference $\|H_i^{(k)} - H_i^{(k-1)}\|$, normalized by a median scale per hop and squashed through a sigmoid so that higher scores indicate smaller changes. A predictive stability variant measures the change in logits produced by fusing up to hop k with a stop-gradient at $k - 1$. A spectral proxy via local Rayleigh quotient drop can be added without assuming homophily.

3.5 BUDGETED FUSION

For each node, we set a mixing budget ρ_i that caps how much risky mass can be aggregated across hops. We then choose fusion weights $w_i^{(k)}$ over hops to maximize a utility score while satisfying the risky-mass budget and the simplex constraints. The utility score $s_i^{(k)}$ combines learnable hop biases, risks, and denoising proxies. A per-node dual variable enforces the budget through a temperature-scaled softmax over $s_i^{(k)}$ minus the dual times $r_i^{(k)}$. Optional hard gating zeroes hops with risk above a threshold κ_i before softmax.

3.6 ASSUMPTIONS AND SCOPE

MiGAD assumes access to decoupled $H^{(k)}$ and tractable estimation of $M^{(k)}$. It is designed as a fusion-stage guard for decoupled backbones Chen et al. (2022) and complements normalization layers [zhao2019_{pairnorm}, zhou2020_{towards}, guo2023_{contranorm}] and depth selectors Zhan et al. (2021). *Analyses of attentive agnostic fusion* [wu2023_{demystifying}, wu2022_{non}].

4 METHOD

We detail MiGAD, a constrained hop-fusion layer that produces a deterministic, interpretable per-node, per-hop mixing policy when plugged into decoupled multi-hop backbones.

4.1 DECOUPLED MULTI-HOP CACHE

Given row-stochastic P and features X , we precompute $H^{(k)} = P^k X$ for k from 0 to L using sparse matrix multiplications. These cached tensors support minibatching and add time proportional to L times the number of edges times the feature dimension.

4.2 COMPACT, IDENTIFIABLE SOFT GROUPS

A lightweight head maps X , or a shallow fusion of the $H^{(k)}$, to logits and applies a row-softmax to obtain $S \in \mathbb{R}^{N \times G}$. We discourage collapse via entropy regularization and anchor interpretability via K-means prototypes on X , optionally aligned to label centroids on labeled nodes. For scalability, we retain only the top- ϱ entries per S_i and renormalize rows, reducing memory from $\mathcal{O}(NLG)$ to $\mathcal{O}(NL\varrho)$.

4.3 LEARNABLE GROUP COMPATIBILITY

We parameterize $C \in \mathbb{R}^{G \times G}$ as a row-stochastic low-rank factorization $C = \text{softmax}_{\text{row}}(UV^\top)$. We warm-start near identity with small off-diagonal noise and, when labels exist, may initialize from class confusion. Mild Kullback-Leibler regularization to the identity and row entropy maintain identifiability while permitting flexibility.

4.4 PER-NODE, PER-HOP MIXING RISK

We diffuse groups as $M^{(k)} = P^k S$. For node i and hop k , the base alignment $a_i^{(k)}$ equals the inner product of $S_i C$ and $M_i^{(k)}$; the base risk is $r_i^{(k)} = 1 - a_i^{(k)}$. Robust alternatives use the Jensen-Shannon divergence between $S_i C$ and $M_i^{(k)}$, scaled to $[0, 1]$, or one minus their cosine similarity. For large graphs, $M_i^{(k)}$ can be estimated by Monte Carlo k -step random walks with an exponential moving average across epochs.

4.5 DENOISING PROXY

We favor hops that stabilize representations by using a feature stability proxy: compute $\|H_i^{(k)} - H_i^{(k-1)}\|$, divide by a median-based scale per hop, and pass through a sigmoid with a temperature that can be annealed. A predictive stability variant uses the change in logits produced by fusing up to hop k with a stop-gradient at $k - 1$. An optional spectral proxy via local Rayleigh quotient drop is compatible with heterophily.

4.6 BUDGETED, RISK-AWARE HOP ATTENTION

We define scores $s_i^{(k)} = \theta_k - \beta r_i^{(k)} - \gamma d_i^{(k)}$, where θ_k are learnable hop biases and β, γ are learned coefficients. We compute weights w_i by solving, for each node, a linear objective in w_i under three constraints: the risky-mass budget $\sum_k w_i^{(k)} r_i^{(k)} \leq \rho_i$, weights lie on the simplex, and weights are nonnegative. The Lagrangian with a per-node dual λ_i leads to logits proportional to $(s_i^{(k)} - \lambda_i r_i^{(k)}) / \tau$, and w_i is the softmax over hops of these logits. We find λ_i by one-dimensional bisection using the monotonicity of risky-mass usage in λ_i . Optional hard gating zeroes hops with risk above κ_i prior to softmax for extra safety.

Algorithm 1 Per-node budgeted hop attention via dual bisection

Require: risks $\{r_i^{(k)}\}_{k=0}^L$, scores $\{s_i^{(k)}\}_{k=0}^L$, budget ρ_i , temperature τ , tolerance ϵ , max iters T , large Λ

- 1: $\lambda^{\text{lo}} \leftarrow 0, \lambda^{\text{hi}} \leftarrow \Lambda$
- 2: **for** $t = 1$ to T **do**
- 3: $\lambda \leftarrow (\lambda^{\text{lo}} + \lambda^{\text{hi}})/2$
- 4: $\alpha_k \leftarrow (s_i^{(k)} - \lambda r_i^{(k)})/\tau$ for all $k \in \{0, \dots, L\}$
- 5: $w_k \leftarrow \exp(\alpha_k) / \sum_{j=0}^L \exp(\alpha_j)$
- 6: $m(\lambda) \leftarrow \sum_{k=0}^L w_k r_i^{(k)}$
- 7: **if** $|m(\lambda) - \rho_i| \leq \epsilon$ **then**
- 8: **break**
- 9: **else if** $m(\lambda) > \rho_i$ **then**
- 10: $\lambda^{\text{lo}} \leftarrow \lambda$
- 11: **else**
- 12: $\lambda^{\text{hi}} \leftarrow \lambda$
- 13: **end if**
- 14: **end for**
- 15: **return** $\{w_k\}_{k=0}^L$

4.7 ADAPTIVE MIXING BUDGET

We set $\rho_i = \rho_0 \cdot b_i$, where b_i reflects uncertainty or boundaryness. One instance uses b_i equal to a convex combination of $1 - \max_g S_{ig}$ and the average of $1 - S_i \cdot S_j$ over 1-hop neighbors, scaled into $[\rho_{\min}, \rho_{\max}]$. The scalar ρ_0 is tuned on validation or calibrated on a sketch so that empirical denoising-to-mixing ratios remain at least one at selected depths, aligning with non-asymptotic trends Wu et al. (2022).

4.8 REPRESENTATION AND PREDICTOR

We fuse Z_i as $\sum_k w_i^{(k)} f_k(H_i^{(k)})$, with either a shared or hop-specific linear map f_k , and add a residual from X . Any normalization layer such as PairNorm, DGN, or ContraNorm can be applied to Z and remains complementary [zhao2019_{pairnorm}, zhou2020_{towards}, guo2023_{contranorm}]. A linear head maps Z_i plus X to class logits.

4.9 OBJECTIVE AND SCHEDULE

The loss combines the task loss with a mixing penalty on budget violation: the average over nodes of the squared positive part of $\sum_k w_i^{(k)} r_i^{(k)} - \rho_i$, multiplied by a coefficient, plus small regularizers on S and C . We use a two-phase schedule: warm up with loose budgets (larger ρ_0 , smaller constraint weight, larger temperature) to stabilize S , C , and f_k , then anneal to tighter budgets and smaller temperature to induce sparse hop selection. Early stopping uses validation loss; we cache hop weights, risks, and budgets for deterministic inference.

4.10 INTEGRATION AND SCALING

MiGAD replaces the fusion stage in SIGN, SGC, and GPR, and the hop attention in hop-interaction models Chen et al. (2022). In APPNP, we tie hop biases to personalized PageRank coefficients while retaining the budgeted gate. For large graphs, we combine top- ϱ sparsity for S , random-walk estimation of $M^{(k)}$, subsampling nodes for dual updates, and exponential moving averages for risks.

4.11 THEORY SKETCH AND DIAGNOSTICS

Under a multi-class block model with unknown groups and compatibility, if S and C approach their true counterparts, $\tau \rightarrow 0$, and the constraint penalty grows large, the learned policy enforces the

risky-mass budget per node. Coupled with a denoising proxy that decreases variance up to a safe depth k^* , the policy certifies aggregation only where denoising dominates mixing, reducing expected cross-group mass relative to risk-agnostic fusion. We diagnose behavior via per-node hop weights, budget usage, learned C heatmaps, risk profiles across hops, inference-time budget sweeps, and safety audits of nodes that consume most of their budgets.

5 EXPERIMENTAL SETUP

5.1 IMPLEMENTATION AND RELIABILITY

We implement MiGAD in PyTorch with PyTorch Geometric sparse operators. Dependencies include torch, torch_geometric, torch_scatter, torch_sparse, numpy, scipy, scikit-learn for K-means, optuna for lightweight hyperparameter search, and pandas. Determinism is enforced by setting seeds for Python, NumPy, and PyTorch; enabling deterministic algorithms where supported; disabling cuDNN benchmarking; and logging library and CUDA versions. We adhere to standard public splits for Planetoid and OGB. For heterophily datasets without standard splits, we perform 10 stratified random splits. We report mean and standard deviation across splits and provide 95% confidence intervals using the t-distribution. For stochastic baselines, we run multiple seeds per split to separate training randomness from inference determinism.

5.2 DATASETS

Homophilous: Cora, Citeseer, PubMed (Planetoid) and optionally ogbn-arxiv. Synthetic: stochastic block models with 5 classes, node counts such as 10k and 50k, within-class probabilities around 0.07-0.10, cross-class probabilities around 0.005-0.01, and class-conditioned Gaussian features with signal-to-noise ratios in 1.0-3.0. Heterophilous: Chameleon, Squirrel, Actor, and WebKB (Cornell, Texas, Wisconsin). Corruptions include edge rewiring toward cross-label edges at rates 0.0-0.3, additive feature noise at 0.0-0.2 times per-dimension standard deviation, and 20% label flips on training nodes.

5.3 BASELINES

We include SIGN with uniform fusion and with unconstrained hop attention, SGC, APPNP with fixed personalized PageRank coefficients, GPR-GNN with learned PPR coefficients, NDLS for node-dependent depth, and normalization layers PairNorm, DGN, and ContraNorm applied after fusion [zhao2019_{pairnorm}, zhou2020_{towards}, guo2023_{contranorm}, zhang2021_{node}]. Stochastic baselines are DropEdge and D

5.4 MiGAD CONFIGURATION

We precompute $H^{(k)} = P^k X$ with row-stochastic P for depths 2, 4, 8, and 12. We use 8 or 16 groups, enforce top- $q = 4$ nonzeros per row in S , and compute risk via alignment for large real graphs and via Jensen-Shannon divergence for synthetic block models. The denoising proxy uses feature stability with a temperature schedule. Budgets derive from uncertainty features with a convex combination parameter in $[0, 1]$, scaled into $[\rho_{\min}, \rho_{\max}]$. The constrained hop attention employs per-node bisection over the dual λ_i with temperature annealing. Hard gating with a threshold around 0.9 activates after warmup.

5.5 TRAINING PROTOCOL

We optimize with Adam using learning rates from $1e-3$ to $3e-3$ and weight decay from $1e-4$ to $5e-4$. Early stopping tracks validation loss with patience between 100 and 300 and a maximum of 2000 epochs for small graphs. Warmup lasts roughly 300-400 epochs with loose budgets (larger ρ_0 , smaller budget penalty, larger temperature), followed by tightening. Hyperparameters are tuned on validation using small Optuna sweeps over ρ_0 , group count, learning rate, and weight decay; test labels remain unseen. We record per-epoch accuracy, loss, constraint usage, and runtime; we persist checkpoints and diagnostic artifacts including learned compatibility heatmaps, budget usage histograms, per-hop weight distributions, oversmoothing curves, and budget sweep curves.

5.6 EVALUATION METRICS

Primary metrics are test accuracy and Macro-F1. For depth robustness, we compute the area under the accuracy-depth curve across the set of depths. For oversmoothing, we track cosine-to-mean and embedding energy across hops. For calibration, we compute expected calibration error with 15 bins. For efficiency, we record epoch time and peak GPU memory. For constraints, we monitor the average risky-mass usage $\sum_k w_i^{(k)} r_i^{(k)}$ relative to ρ_i and the fraction of nodes with active duals ($\lambda_i > 0$). We perform paired statistical tests against a primary baseline (e.g., SIGN with attention) using matched seeds.

5.7 EXPERIMENTS

- **Experiment 1: Depth-robust accuracy and oversmoothing control.** Datasets are Cora, Citeseer, PubMed, optionally ogbn-arxiv, and synthetic block models. We precompute $H^{(k)}$ up to depths 2, 4, 8, and 12. Baselines include SIGN (uniform and attention), SGC, APPNP, GPR-GNN, NDLS, and normalization layers [zhao2019_pairnorm, zhou2020_towards, guo2023_contranorm, zhang2021_node]. We report test accuracy, Macro-F1, area-under-depth, oversmoothing proxies, calibration, efficiency, and constraint adherence. **Experiment 2: Label edges (0.0–0.3), additive feature noise (0.0–0.2), and 20% training label flips.** Metrics are accuracy and Macro-F1 across perturbations, robustness AU C across heterophily rates, inference determinism (standard deviation at inference without retraining to check monotone control of risky mass).
- **Experiment 3: Interpretability, ablations, and counterfactual checks.** Datasets are synthetic block models and one real dataset (Cora or Chameleon). Analyses include correlation between boundaryness (fraction of differently labeled neighbors) and both budget usage and number of gated hops; alignment of learned compatibility with empirical class confusion; and counterfactual enabling of each node’s most risky hop to measure margin changes. Ablations remove the constraint, use uniform S , add label guidance to S , fix C to identity, drop the denoising proxy, and toggle hard gating. We report accuracy deltas, constraint violations, correlation coefficients with p-values, and the fraction of nodes harmed by counterfactual risky-hop enablement.

5.8 NON-APPLICABLE LINES

Universal graph convolution results address expressivity rather than risk-constrained diffusion and are not used as baselines [author_year_niversal]. Inductive knowledge graph completion targets a different supervision regime and graph type [author_year_nuniversal].

6 RESULTS

6.1 ENVIRONMENT EXECUTION

The captured logs show successful installation of PyTorch 2.8 and dependencies and the launch of the experiment runner. The shared console output does not contain numerical outcomes such as accuracies, robustness curves, oversmoothing metrics, or diagnostic aggregates. In line with our policy to report only results present in logs, we therefore do not state quantitative performance.

6.2 PLANNED EVIDENCE ONCE RUNS COMPLETE

The implemented design directly tests MiGAD’s intended properties.

- **Depth robustness and oversmoothing control.** By computing area under the accuracy-depth curve and tracking cosine-to-mean and energy across hops, constrained hop attention is evaluated for sustaining accuracy at larger depths and reducing oversmoothing proxies relative to risk-agnostic fusion, consistent with the local mixing-denoising rationale Wu et al. (2022). Baselines include spectral and hop-decoupled fusions without local constraints [bruna2013_spectral, deferrard2016_convolutional, chen2022_from]. **Heterophily and corruption robustness.** Unlabeled edges and additive feature noise, MiGAD’s budget should bind more frequently, reducing risky—

*massconsumptionandyieldinggracefuldegradationrelativetounconstrainedattention.Stochasticmethodsca
usagecurvesbyconstructionofthegate.*

- **Interpretability and constraint adherence.** We will report per-node dual activity, budget usage relative to ρ_i , and learned compatibility heatmaps. On synthetic block models, Jensen-Shannon risks allow aligning learned compatibilities with known structure; on real graphs, we will correlate compatibility with empirical class confusion. Boundaryness is expected to correlate positively with constraint activity, and counterfactual enabling of the most risky hop should reduce predictive margins for a substantial fraction of nodes, evidencing the guardrail effect.

6.3 FAIRNESS, RELIABILITY, AND EFFICIENCY

Backbones, depths, and parameter counts are matched across MiGAD and baselines. Hyperparameter grids are shared where possible to mitigate selection bias. Early stopping and validation-only tuning prevent test leakage. Efficiency is monitored via epoch time and peak memory; group diffusion and risk estimation add overhead proportional to the number of hops times the number of edges times the group count, mitigated by top- q sparsity and random-walk approximations.

6.4 LIMITATIONS OF CURRENT EVIDENCE

The present logs lack numerical outputs from Experiments 1-3. While the methodology and code are in place, quantitative claims await execution outputs. Additionally, if temperatures or budget scales are poorly tuned, training may become brittle; our warmup and annealing schedule is designed to stabilize learning but should be validated per dataset.

6.5 HOW TO COMPLETE THE EVALUATION

Execute the provided runners for the three experiments, persist CSV logs and artifacts (compatibility heatmaps, budget usage histograms, per-hop weight distributions, over-smoothing curves, budget sweep plots), and report mean and standard deviation with 95% confidence intervals across seeds. Perform paired tests against a primary baseline such as SIGN with attention. Cite baselines and theoretical context appropriately [zhao2019_{pairnorm}, zhou2020_{towards}, guo2023_{contranorm}, rong2019_{dropedge}, papp2021_{dropgnn}, zhang2021_{node}, ch

7 CONCLUSION

We presented MiGAD, a budgeted, compatibility-aware diffusion guard that gates multi-hop aggregation per node and per hop. By learning soft groups and a row-stochastic compatibility matrix, estimating per-node, per-hop mixing risk, and enforcing a node-specific budget via a differentiable constrained attention layer, MiGAD turns local mixing-denoising theory into a practical, deterministic policy for selective diffusion. The module integrates seamlessly with hop-decoupled backbones and remains complementary to normalization and stochastic techniques [chen2022_{from}, zhao2019_{pairnorm}, zhou2020_{towards}, guo2023_{contranorm}, rong2019_{dropedge}, papp2021_{dropgnn}]. *Walkriskestimation, adaptivebudgets, andatwo-phaseschedule, togetherwithrigorousexperimentalprotocolsfordep*

The current execution established the environment and pipeline but produced no numerical outputs in the shared logs; accordingly, we make no quantitative claims here. The specified procedures and diagnostics are designed to produce clear, reproducible evidence once runs are executed: improved depth-accuracy area under curve with constrained oversmoothing, graceful degradation under heterophily and noise, deterministic inference with interpretable budgets, and counterfactual tests validating the guardrail effect. Future work includes extending budgeted-risk gating to graph transformers by budgeting long-range attention, to temporal and heterogeneous graphs via time- and type-conditioned compatibility and budgets, and to governance tasks such as active learning and rewiring curricula that use budget usage as a decision signal. More broadly, our results advocate constraining aggregation-rather than repairing it post hoc-as an interpretable and robust path to deeper graph learning [wu2022_{on}, wu2023_{demystifying}].

This work was generated by AIRAS (Tanaka et al., 2025).

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